Clément LONGEAC

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Final-year ESIEE Paris engineering student (Data Science & AI, Research Track), aiming for a PhD in applied AI. Built DEMNE, a routing framework combining rule cascades, Transformers, and LLMs with per-task threshold calibration and few-shot prompting; co-authored a paper under review; and deployed it in production at AP-HP. Skilled in Python/PyTorch and in evaluating generative models (BioMistral, GPT-4) in clinical settings. **Available for a six-month internship starting mid-February 2027 after graduation.**

Research Interest

Self-supervised representation learning (JEPA-architecture) · Frugal and calibrated inference · Clinical NLP for French medical text · AI for biomedical research

Education

ESIEE Paris — Noisy-le-Grand, France
Engineering degree (five-year program, equivalent to an MSc in Engineering in Data Science & AI major) + Research Program (Advisor: Akram Redjdal); Deep Learning focus (year-long project) — *Sep. 2024 – Jul. 2027*

Politechnika Lubelska (Lublin University of Technology) — Lublin, Poland
ERASMUS semester — Faculty of Electrical Engineering and Computer Science — *Sep. 2026 – Feb. 2027*

Lycée Claude Fauriel (CPGE) — Saint-Étienne, France
Two-year post-baccalaureate intensive preparatory program (Classes Préparatoires) in Engineering Sciences / Physics and Chemistry; specialization in Physics and Chemistry — *Sep. 2021 – Jul. 2024*

Experience

Research Intern (Research Track Program) — Nov. 2025 – Aug. 2026
ESIEE Paris | LBA (Aix-Marseille Univ.) | AP-HP (EDS) — *Paris, France*

- Designed **DEMNE**, a three-tier adaptive routing framework assigning each task to the most frugal paradigm (rules / transformers / LLM) under a calibrated performance threshold (applied to clinical entity extraction)
- Developed 5 corpus-derived metrics computed automatically, eliminating the need for task-specific feature engineering
- External validation on 300 AP-HP clinical reports (colorectal, ENT): $F_1 = 0.941$ (rules tier); 93.7% of entities processed without GPU
- Estimated 3–4 orders-of-magnitude reduction in energy consumption vs. a 100% LLM pipeline; framework aligned with the EU AI Act
- Open-sourced the project (metrics pipeline + Streamlit dashboard); Supervisor: Dr. Akram Redjdal; Co-supervisor: Prof. E. Kempf

Software Engineering Intern (GPU/CI) — May 2025 – Aug. 2025
Synchrotron SOLEIL — *Paris-Saclay, France*

- Wrote `autopkgtest` scripts in OpenCL to validate software compatibility on AMD GPUs, executed on the Debian ROCm CI infrastructure (GitLab Salsa)
- Automated test procedures (install/uninstall) via `rocm-test-launcher` on clean VMs
- Goal: map AMD GPU scientific-compatibility to improve ROCm support in Debian

Private Tutor (Math & Physics) | Complétude — Dec. 2024 – May 2026
Part-time — *France*

Volunteer | La Main Tendue pour le Pérou — 2020 – 2022
Non-profit association — *Neuville-sur-Saône, France*

Publications

[1] **Clément LONGEAC***, Akram Redjdal, Clément Arbellot de Vacqueur, Emmanuelle Kempf (2026) (under review). "DEMNE: Adaptive Selection of Named Entity Extraction Method for Frugal Clinical NLP in Oncology". *Methods of Information in Medicine*.

Technical Skills

Systems: Linux/Debian, Windows, Android
Programming: Python, C, C++, Bash/Shell, SQL, VHDL
ML & NLP: PyTorch, TensorFlow, Hugging Face Transformers, spaCy, scikit-learn
LLMs & Research: DrBERT, BioMistral, GPT-4, Claude API, fine-tuning, few-shot prompting
Clinical NLP: EDS-NLP (`edsnlp`), BRAT annotation, CoNLL NER, `eco2ai`
Methods: grid search optimization, asymmetric loss, corpus analysis, frugal AI,Fine-tuning
Tools & DevOps: AWS, Docker, Kubernetes (K8s), Git, GitHub, GitLab, CI/CD, Streamlit, Pandas
Office: Microsoft Office, LibreOffice, Google Workspace
Compliance: EU AI Act (Art. 12/14), GDPR, HDS, AFNOR SPEC 2314
GPU: AMD ROCm, OpenCL

Languages

French (native) – English (Professional working proficiency) – Spanish (B1) – Japanese (A1) – Italian (A1)

Licenses & Certifications

PIX certification *obtained* – Voltaire certification *obtained*

Soft Skills

Research mindset and scientific rigor – Proactive problem solving (DEMNE) – Autonomy (research) – Teaching and pedagogy (Complétude) – Teamwork (AP-HP, Synchrotron SOLEIL) – Adaptability (ERASMUS, Synchrotron SOLEIL) – Leadership (Synchrotron SOLEIL)

Interests

AI safety & alignment – Basketball (competitive) – Beekeeping – Particle physics (CERN, ITER) – Science communication (Carl Sagan, Stephen Hawking) – Chess

References

Research Supervisor: Dr. Akram Redjdal (akram.redjdal@esiee.fr)

Internship Director: Emmanuel Farhi (emmanuel.farhi@synchrotron-soleil.fr)

Frédéric-Emmanuel Picca (frederic-emmanuel.picca@synchrotron-soleil.fr)